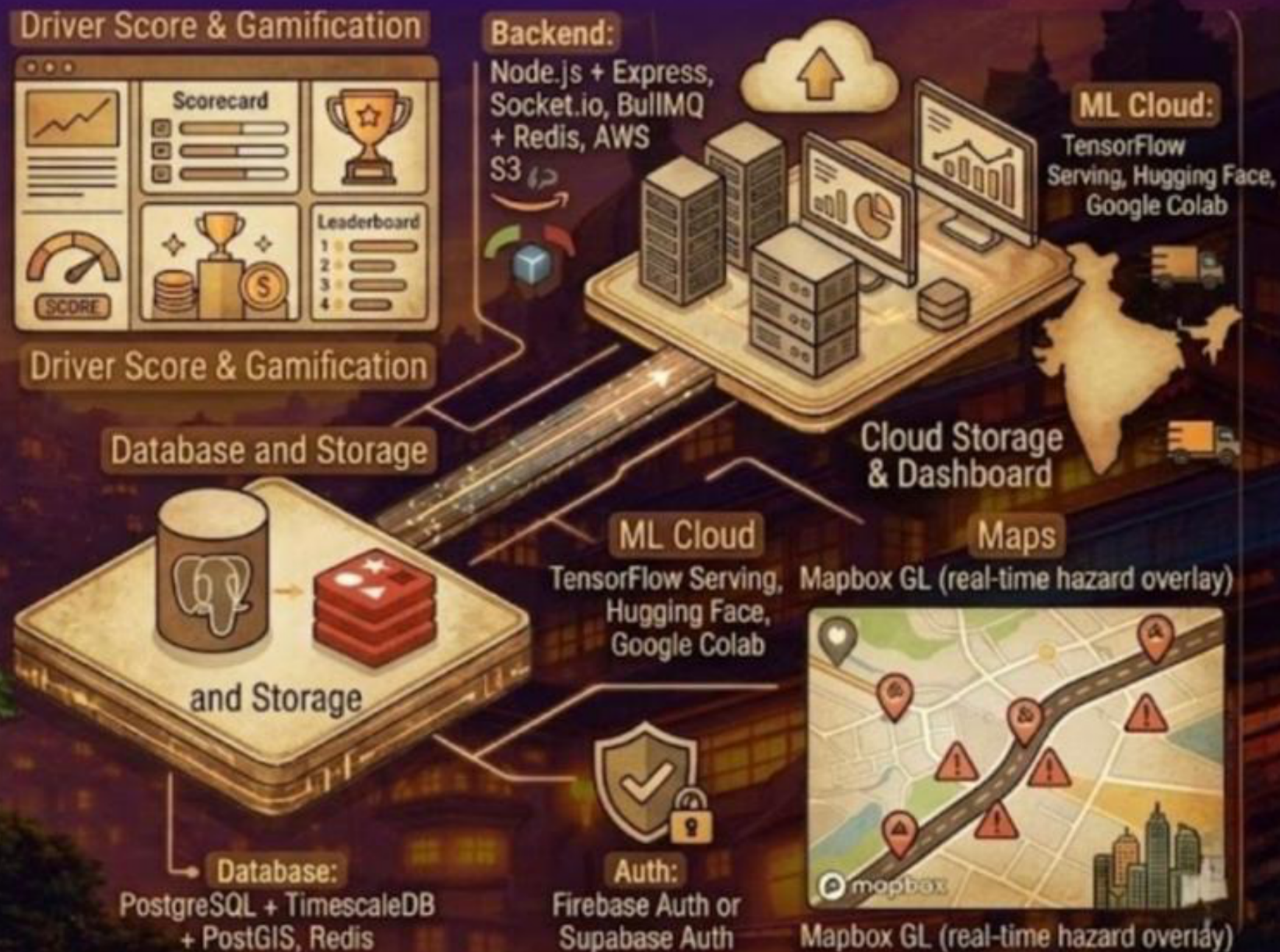


Technical Approach

Detailed Tech Stack		
	Layer	Tech Stack
	Frontend	React Native (Expo), Zustand, Expo Sensors, Socket.io client
	Backend	Node.js + Express, Socket.io, BullMQ + Redis, AWS S3
	ML On-device	TensorFlow Lite, MediaPipe, YOLOv8-tiny
	ML Cloud	TensorFlow Serving, Hugging Face, Google Colab
	Database	PostgreSQL + TimescaleDB + PostGIS, Redis
	Auth	Firebase Auth or Supabase Auth
	DevOps	Docker, Railway/Fly.io, GitHub Actions
	Maps	Mapbox GL (real-time hazard overlay)



System Architecture